

THE EMERGENCE OF COMPLEX BEHAVIOR IN LARGE-SCALE ECOLOGICAL ENVIRONMENTS

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ABSTRACT

We examine how environmental extent, population size, and sensing structure shape the emergence of complex behavior in open-ended ecological worlds. In this setting, agents receive no explicit rewards, supervision, or externally specified learning objective. Their neural network policies instead evolve through reproduction, mutation, and selection as agents compete for the resources needed to survive. Because agents both act within and modify a shared environment, behavioral adaptation and environmental change form a coupled ecological process. The study therefore does not seek a single optimized policy; it establishes how behavioral capabilities arise, persist, and vary in large evolving populations subjected to competitive and environmental pressure. Using modern accelerators and a new multi-agent simulator, we scale ecological evolution substantially beyond prior experiments, reaching populations of more than 60,000 agents, each with its own evolved neural network policy. Across these worlds, we identify emergent long-range resource extraction, vision-guided foraging, and predation. The experiments further show that sensing modalities and environmental scale are not incidental implementation details: they determine whether these behaviors emerge, how reliably they recur across seeds, and how stable they remain over evolutionary time. Several behaviors appear only in sufficiently large environments and populations, and larger scales make the observed ecological dynamics more consistent. These results establish large-scale ecological simulation as a concrete experimental instrument for studying machine learning systems in which adaptive behavior is produced by survival, reproduction, and environmental interaction rather than by explicit task optimization. Experimental code will be released upon acceptance.

1 INTRODUCTION

Since Darwin, the evolutionary emergence of biodiversity and complex behavior has been a central explanatory problem in evolutionary theory (Darwin, 1859; Simpson, 1944), ecology (MacArthur & Wilson, 2001), and genetics (Dobzhansky, 1937). Formal and empirical accounts of these dynamics span theories of speciation and adaptive radiation (Coyne & Orr, 2004; Schluter, 2000) as well as mathematical analyses of population genetics (Fisher, 1930; Wright, 1931). Together, this literature establishes evolution as a mechanism capable of generating organized behavior from local variation, selection, and ecological interaction.

This scientific tradition has produced substantial insight through mathematical modeling, laboratory experimentation, and field observation, but direct experimental analysis of emergent behavior in natural ecosystems remains intrinsically constrained. As ecosystems grow in spatial extent, population size, and interaction complexity, they become increasingly difficult to control, instrument, and measure. Even when intervention is technically feasible, controlled experimentation in large natural systems can damage habitats or displace wild populations, limiting the range of ecological hypotheses that can be tested directly.

We study the emergence of complex behavior under competitive and environmental pressure by evolving neural network policies in large-scale ecological simulations. Agents in these simulations are not trained against a prescribed reward function; they acquire resources in order to survive and reproduce, and policy variation enters through mutation when a parent produces offspring. This formulation makes ecological scale an experimentally manipulable variable. It allows us to examine

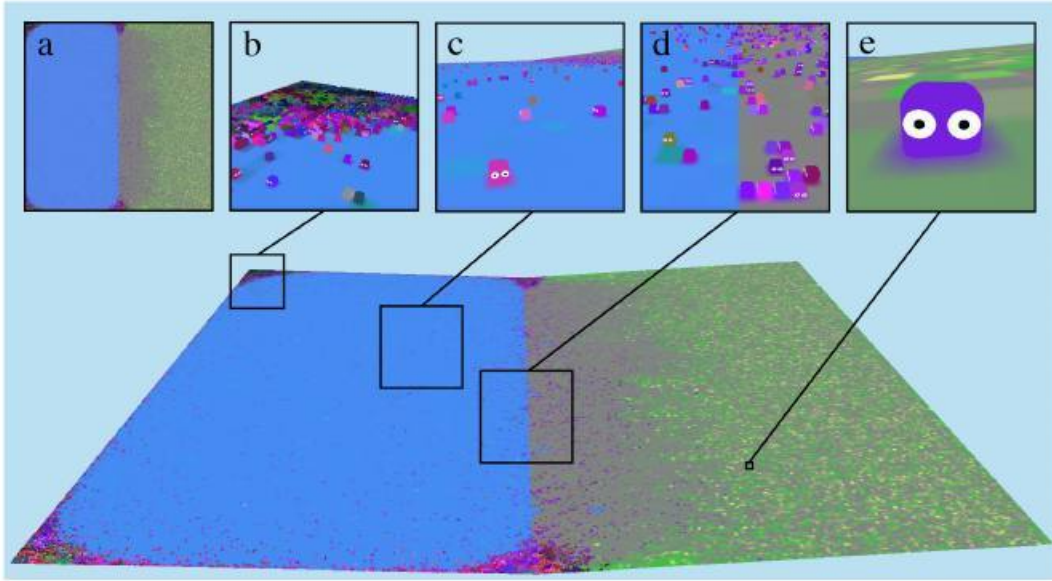


Figure 1: 3D visualization of a 512×512 map in our environment. Insert (a) shows a top-down view of the full map; (b) shows agents clustered near a corner; (c) shows agents in water; (d) shows agents traveling between land and water; and (e) shows a close view of a single agent gathering resources.

evolutionary dynamics in populations of simple digital organisms while isolating how environmental size, population size, and sensing structure influence behavioral emergence.

To conduct this study, we use a new JAX-based simulation environment for rapidly evaluating large grid worlds. The largest experiments contain more than 60,000 individual agents and span more than 1,000,000 grid cells. Within these worlds, agents navigate to obtain food and other resources, and reproduction occurs when an agent has accumulated sufficient resources to create a mutated copy of itself. Figure 1 provides a 3D visualization of the environment and the agent–resource interactions studied throughout the paper.

Our objective is not only to use simulation as a controlled model of ecological phenomena, but also to treat ecology itself as a mechanism for producing machine intelligence. Recent work in AI has shown that scale can transform system capabilities rather than merely improve performance (Wei et al. 2022a). Yet embodied learning studies are often conducted in comparatively small environments with limited population sizes. This paper asks what changes when scale is applied not to a single model, but to an entire evolving ecology. Analogous to the way reasoning abilities can appear only beyond particular model-size regimes (Wei et al. 2022b), we examine which population-level strategies become accessible when environmental extent and ecological complexity are sufficiently large.

Our experiments establish that sensor configuration and environmental scale decisively shape the behaviors produced by open-ended ecological evolution. Agents equipped with simple compass sensors evolve long-range resource-gathering expeditions, whereas the same behavior is substantially less reliable in smaller worlds. Agents equipped with visual sensors evolve more effective foraging and attacking behaviors than agents without vision, and these advantages are clearer and more stable at larger scales. Together, these results demonstrate that modern hardware accelerators make it possible to study ecological evolution at scales where rich behavior emerges from survival pressure alone. The paper contributes a scalable experimental setting for objective-free ecological evolution and uses it to show that scale and sensing are central determinants of emergent adaptive behavior in artificial ecologies.

2 RELATED WORK

Computational models of life and evolution have shaped artificial intelligence since the field’s early history. Conway’s Game of Life (Games 1970) established that simple local transition rules can generate coherent population-level structure, while Tierra (Ray 1992) and Avida (Brown) modeled life as self-replicating programs that evolve through mutation. In the same local-rule tradition, Reynolds (1987) showed that flocking and schooling can emerge from compact motion rules rather than explicit global coordination.

Evolutionary principles have also been used to generate agent bodies and controllers, including locomotion morphology (Sims 1994a,b; Ha 2018; Gupta et al. 2021; Auerbach et al. 2014; Auerbach & Bongard 2012; Jeon et al. 2025; Auerbach & Bongard 2014; Szerlip & Stanley 2013) and sensing systems (Pratt et al. 2022; Mugan & MacIver 2020; Tiwary et al. 2025) for virtual agents, while neuroevolution methods directly evolve neural network architectures (Stanley & Miikkulainen 2002, 2004; Stanley et al. 2009). Our study uses this evolutionary foundation for a different purpose: rather than optimizing agents against a reward or objective signal, we examine which behaviors arise from ecological pressure itself and how those behaviors depend on environmental scale and complexity.

Evolution has also served as a design principle for optimization algorithms. Genetic algorithms (Holland 1992; Jong 1975) and evolutionary strategies (Wierstra et al. 2014; Hansen & Ostermeier 2001; Rechenberg 1973; Schwefel 1995; Conti et al. 2018) use mutation, selection, and population-level search as optimization mechanisms. Related methods encourage novelty among solutions (Lehman & Stanley 2011; Cully et al. 2015), and evolutionary approaches have optimized neural networks for domains such as Atari and Mujoco locomotion tasks (Salimans et al. 2017; Such et al. 2017; Mania et al. 2018). Although our agents are updated through evolutionary mechanisms, the contribution of this paper is neither a new optimizer nor the use of evolution to solve a fixed benchmark. The central object is emergent behavior in an evolving ecology without an externally specified task objective.

A closely related line of work uses biological inspiration to construct artificial organisms within simulated environments. Early examples include Bedau (1992), Yeager et al. (1994), Channon & Damper (1998), and Tisue et al. (2004). More recent systems use cellular automata or artificial-life environments to produce emergent morphologies and behaviors (Mordvintsev et al. 2020; Robinson 2021; Heinemann 2021; Randazzo & Mordvintsev 2023; Caussan 2025). Mordatch & Abbeel (2018) and Park et al. (2023) examine the emergence of language and communication in simulated environments, and autocurriculum research studies how interacting agents can create increasingly challenging learning landscapes for one another (Leibo et al. 2019, 2018; Team et al. 2021). Complementary work by Wang et al. (2019) and Aki et al. (2024) evolves environments optimized for agent learning. These studies establish that environmental structure can elicit sophisticated behavior; our work extends this perspective by directly testing how ecological scale and sensory access affect behavior in large evolving populations.

The most directly comparable studies simulate populations of agents interacting in complex environments. Baker et al. (2019) examine a Hide-and-Seek task in which a small number of hider agents learn to use environmental tools to evade seeker agents, with both sides trained by reinforcement learning. Other work trains populations for specified tasks such as playing video games (Jaderberg et al. 2019), manipulating objects (Petrenko et al. 2021), or capturing agents (Zheng et al. 2017; Yamada et al. 2020). These systems demonstrate important forms of emergence, but they typically remain organized around fixed tasks and objectives. Neural MMO (Suarez et al. 2019, 2023) moves closer to open-ended reinforcement learning by placing agents in a massively multi-agent game and optimizing survival rather than a narrow task, but its population is capped at a fixed number of concurrent agents, with experiments up to 128 agents, and it does not include generational inheritance. Hamon et al. (2023) is closer to our setting because it includes variable population sizes, genetically inspired evolution, and generational timelines, but its environment is substantially smaller, less feature-rich, and not designed to analyze scale directly. Similarly, Lu et al. (2024) simulates agents that evolve to gather resources. However, that work operates at a smaller scale, with a maximum of 256 agents, and emphasizes tool use through programmable robots. In contrast, our experiments study populations that can grow and shrink endogenously, evolve across generations, and interact in worlds large enough to expose scale-dependent behavioral regimes.

Finally, recent high-throughput simulators show how optimized environments can increase training speed and agent count in game-like domains (Koyamada et al. 2023; Matthews et al. 2024; Lechner et al. 2023). Our work draws on this computational orientation but applies it to objective-free ecological evolution, where the relevant outcome is not benchmark performance but the emergence, stability, and population-level consequences of behavior.

3 METHOD

3.1 ECOLOGICAL GAMES

We formalize the interaction between evolving agent populations and their environment as a multi-player dynamical system that we call an **Ecological Game (EG)**. As in Markov decision processes (MDPs) and stochastic games (SGs) used in reinforcement learning and multi-agent reinforcement learning (see Sutton et al. (1998) and Albrecht et al. (2024)), an EG has Markovian transition dynamics. Unlike these objective-driven formalisms, however, the EG is designed to represent changing populations in a reward-free setting. Like an SG, an EG contains a set of agents I , a state space S , an initial state distribution ρ_0 , an action space A , and a transition function $\mathcal{T}$ mapping states s_t and the actions selected by each player $a_{t,i}$ to a distribution over subsequent states s_{t+1} . An EG removes the reward function and instead introduces a population function $\Gamma(s_t)$, which records which agents in I are alive and identifies their parents so that birth and death events can be recovered from state. As MDPs and SGs have partially observable variants, POMDPs and POSGs, an EG can also be partially observable; we refer to this case as a POEG. A POEG differs from an EG by restricting each agent from observing the full state and instead using an observation function $O(s_t, i)$ that maps a state and player to that player’s observation. Our environment is a POEG whose observation function varies with the sensor configuration described below.

3.2 ENVIRONMENT

The POEG instantiated in our experiments is a large grid world in which agents must acquire multiple resources to survive and reproduce. The environment state is factored as $s_t = \{s_t^m, s_t^a\} \in S = S^m \times S^a$, where s_t^m contains map-level two-dimensional fields such as terrain and resource distributions, and s_t^a contains agent-level variables such as position, health, and internal resources.

Terrain: Each map contains a rock layer specifying the height of every grid cell. Water is initially placed on cells whose rock value lies below a preset sea level and then flows downhill during an episode according to a discrete flow simulation defined by the rock layer. The experiments use the six global terrain configurations shown in Figure 2: **Ocean**, **Beach**, **Island**, **Lake**, **Isthmus**, and **Channel**. We evaluate grid sizes of 64×64 , 128×128 , 256×256 , 512×512 , and 1024×1024 . When grid size changes, the initial population is scaled proportionally to map area; after initialization, population size is determined endogenously by survival and reproduction dynamics.

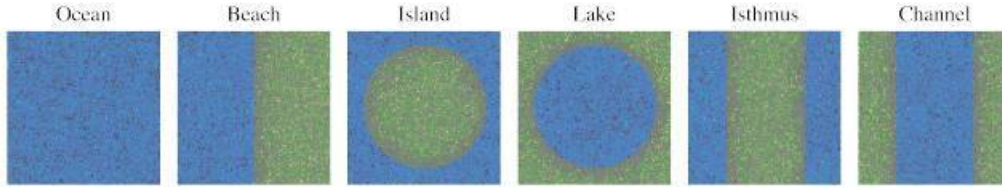


Figure 2: Overhead views of the terrain configurations used in our experiments at 256×256 grid size. Small dark spots denote individual agents, and green and yellow regions denote concentrations of energy and biomass. Each image was captured 5000 steps into an episode, after most water biomass had been consumed but before agents had evolved longer-term survival on land.

Resources and Agent Health: Agents must collect **water**, **energy**, and **biomass** to survive and reproduce. Biomass and energy are initially distributed randomly throughout the environment, and individual agents can subsequently consume and redistribute both quantities.

Every agent action incurs small energy and water costs. Spent water returns to the landscape and flows downhill toward the nearest body of water, whereas spent energy is destroyed. Biomass pro-

vides the material basis for reproduction: each agent requires a fixed amount of biomass to exist and can reproduce only after collecting enough excess biomass to allocate to an offspring. Total biomass, like total water, is conserved during the simulation, but biomass does not move autonomously. Energy is not conserved; instead, it slowly grows on free-standing biomass. These coupled resource dynamics create a more structured ecological problem than single-food-source collection. Appendix F.2 reports ablations on the resource rules.

Agents also maintain health points (HP). Overspending a critical resource causes health to deteriorate rapidly, while available energy and water can be converted into gradual HP recovery. Agents may attack other agents, thereby reducing the targets' HP, and they also receive a small age-dependent damage term that increases over time, so death eventually occurs from old age if no other cause intervenes. When an agent's HP reaches zero, the resources it carries are returned to the grid cell it occupies. Consequently, an agent that kills another agent can consume the released resources, producing a basic form of predation. Appendix F.8 reports ablations on HP hyperparameters.

Sensors: Agents observe the world through four sensor types. The **internal** sensor reports age, health, and carried resources; the **external** sensor reports external resources in the agent's current grid cell; the **compass**, inspired by magnetoreception in birds and other animals, reports global orientation as a 4-way one-hot vector; and **vision** returns a 7×7 top-down rendered patch centered around the agent, with three color channels and a fourth channel encoding local elevation difference. Appendix F.6 reports ablations on vision configuration.

Actions: Agents select from a finite discrete action set organized into five categories. **Rest** performs no external operation and has the lowest energy cost. The three **Move** actions rotate clockwise, rotate counterclockwise, or move forward by one grid cell. Because cells cannot contain multiple agents, attempted moves into occupied cells and simultaneous conflicts for the same destination cell are canceled. A single **Attack** action kills all agents in the 3×3 square directly in front of the attacker; Appendix F.7 evaluates alternative attack strengths. Three **Eat** actions correspond to water, energy, and biomass. If the selected resource is present in the agent's current cell, the action transfers a fixed quantity from the environment into the agent's finite-capacity stomach.

The final action is **Reproduce**, which creates an offspring by single-parent reproduction. Reproduction succeeds only when the parent has accumulated sufficient resources to allocate to the child. The child is placed directly behind the parent, immediately inherits a fixed amount of the parent's resources, and receives a mutated copy of the parent's policy weights.

Several experiments disable specific sensors or the attack action in order to measure the ecological consequences of different agent capabilities. We use three sensing configurations: internal and external resource sensors only (**R**), resource sensors plus compass (**RC**), and resource sensors plus compass and vision (**RCV**). The suffix **+A** denotes that attacking is enabled; for example, (**RC+A**) agents have resource sensors, a compass, and the attack action.

3.3 POLICIES

Agent policies are compact memoryless MLPs. We use memoryless policies to keep the computational model simple and to limit the number of parameters maintained per agent. Raw sensor values are flattened, normalized to $[-1, 1]$, and passed through separate linear layers with output dimension 64. These sensor embeddings are summed and processed by a 2-layer MLP with hidden dimension 64 and ReLU nonlinearities. Because the action space is discrete, the final layer is linear with output dimension equal to the number of available actions; actions are sampled from a softmax distribution whose temperature evolves with the network weights. Weights are represented as half-precision bfloat16 values to support larger populations and faster simulation. Depending on the active sensors, each agent network contains roughly 10k–25k parameters. Appendix F.5 reports ablations on network size and architecture.

At initialization, all agent weights are sampled using Kaiming initialization (He et al., 2015), and all bias vectors are set to zero. Policies for agents born during an episode are generated by copying the parent's weights and adding Gaussian noise with standard deviation 3×10^{-2} to all mutable weight values. This mutation scheme follows the broad approach used in neuroevolution research (Such et al., 2017). Although more elaborate mechanisms such as crossover are possible, we use single-

parent mutation to isolate the effect of environmental scale within a deliberately simple evolutionary setting. Empirically, the use of low-precision floating point values required a larger mutation rate; Appendix F.4 reports the corresponding ablation.

Agents also carry a 3-channel trait that determines their rendered color on the map. This color trait mutates in the same manner as the network weights and determines the agent’s appearance to other agents with vision. In principle, this trait permits visual agents to distinguish among different types of agents.

3.4 COMPUTATION AND REPRODUCIBILITY

All experiments are implemented in JAX (Bradbury et al. 2018) and run on Nvidia H100 and A40 GPUs using a state-of-the-art simulation environment that evaluates environment dynamics, agent policies, and measurements on modern accelerators. Unless otherwise noted, each run proceeds for 2M environment steps or until population extinction. Per-seed wall-clock time ranges from less than one hour for 64×64 grids to approximately 10 hours for 1024×1024 grids on a single H100 GPU. Appendix E describes the simulator architecture, reports additional scaling measurements, and specifies reproducibility considerations.

4 EXPERIMENTS

The experiments evaluate how sensing modality, environmental scale, and population size govern the behaviors that emerge from open-ended ecological evolution. We organize the empirical analysis around two regimes in which the relevant behavioral claims can be directly measured: long-distance resource gathering in structured terrestrial terrains, and visual foraging and predation in open-water environments.

4.1 LONG-DISTANCE RESOURCE GATHERING

The first experiment tests whether agents evolve the ability to travel long distances inland to obtain resources and then return to water. In terrains with clear directional structure, agents with a compass (**RC**) develop substantially more effective long-range resource-gathering behavior than the baseline agents without a compass (**R**). The same experiments show that this behavior becomes more consistent and ecologically consequential as environmental scale increases, linking individual navigation to population-level persistence.

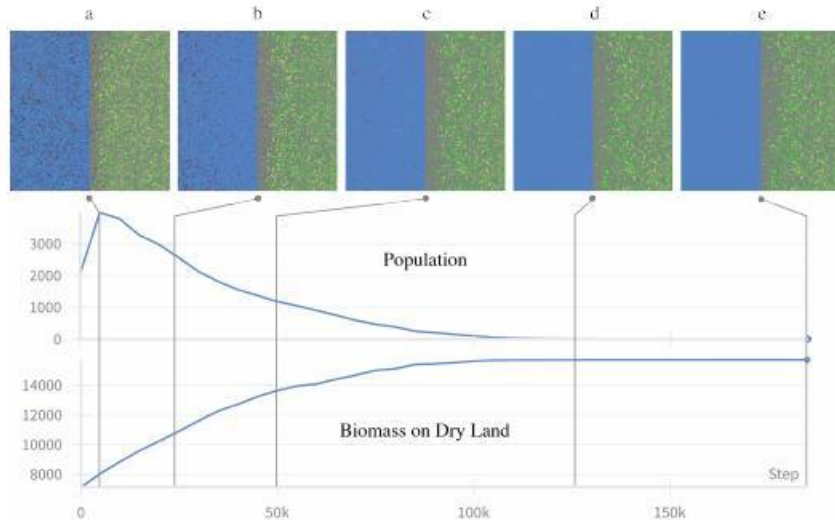


Figure 3: Representative trial with (**R**) agents in a 256×256 **Beach** environment. The top panel reports population over time; the bottom panel reports free biomass on dry land.

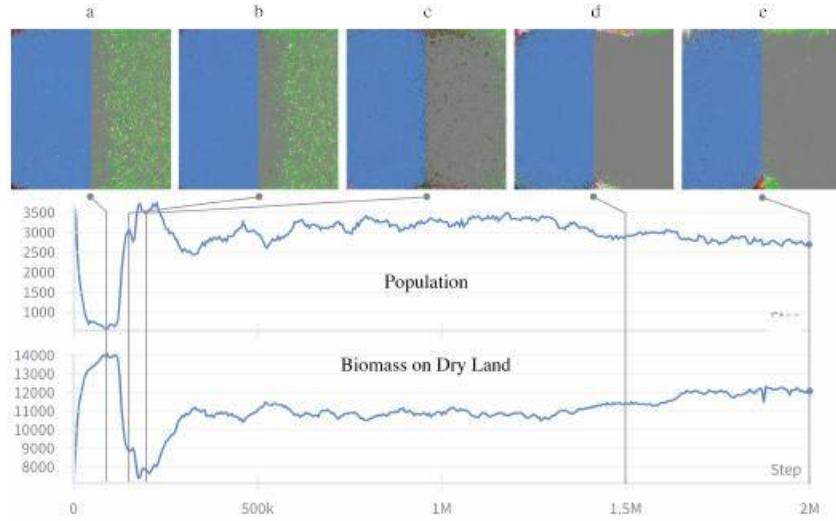


Figure 4: Representative trial with **(RC)** agents in a 256×256 **Beach** environment. The top panel reports population over time; the bottom panel reports free biomass on dry land.

Figure 3 illustrates the baseline **(R)** condition. After 5000 time steps (a), the initial agents on dry land have died from lack of water, while agents in the water have rapidly expanded because of abundant initial resources. At this stage, the population has not yet produced policies that reliably seek out and return from inland resources. From the early population peak, agents in the water occasionally move onto land (b), where they often fail to return to the ocean and die, depositing their biomass on land. By 50,000 steps (c), this gradual transfer has nearly doubled the uneaten biomass on land. Because biomass has moved from water to land, the water environment can no longer support a large population. By 125,000 steps (d), almost all biomass has accumulated on land and the population is nearly extinct; the final agent dies at approximately 185,000 steps (e).

This trajectory demonstrates that local navigational failure can induce population-level ecological collapse. Individual deaths change the spatial distribution of resources, and that redistribution feeds back onto the survival prospects of the entire population. The same redistribution also creates a selective opportunity: any agent that can travel onto land, exploit the biomass deposited by previous generations, and return safely to water can access a large resource reserve unavailable to agents without reliable long-distance navigation.

Figure 4 shows the corresponding condition with compass-equipped **(RC)** agents. The run initially follows the same ecological pattern as the baseline: the population contracts while free biomass on land increases (a). However, some agents use the compass to drift into the relative safety of the corners. Around 150,000 steps (b), agents begin to “mine” land biomass by navigating between land and water. By 200,000 steps (c), these mining agents have cleared the beach. The population subsequently changes in local organization, including the formation of clumps near edges (d), but remains stable through the full 2,000,000-step run (e).

We quantify this phenomenon across **Beach** worlds with grid sizes 1024×1024 , 512×512 , 256×256 , 128×128 , and 64×64 , initialized with 32768, 8192, 2048, 512, and 128 agents, respectively. Each configuration is evaluated with four random seeds. We record whether mining emerges, defined as a drop of at least 10% in free dry-land biomass from its historical maximum, and also record the onset time of the mining event, defined by the local maximum in land biomass, together with the subsequent percentage drop. Table 1 summarizes mining and extinction events, and Table 3 in Appendix A gives seed-level data. Across these comparisons, mining emerges substantially more reliably in **(RC)** populations than in **(R)** populations. It appears in every **(RC)** seed at and above the 256×256 scale, whereas smaller environments show less stable populations and higher early-extinction rates.

Table 1: Mining events in the **Beach** world in populations of agents with a compass (**RC**) compared to populations without a compass (**R**) across five world sizes and initial populations.  denotes the number of seeds that featured a mining event.  denotes the number of seeds in which the population went extinct before 2M steps. See Table 3 in Appendix A for detailed data from each seed.

	1024 × 1024		512 × 512		256 × 256		128 × 128		64 × 64	
	32768		8192		2048		512		128	
										
(RC)	4/4	0/4	4/4	0/4	4/4	0/4	3/4	1/4	2/4	3/4
(R)	1/4	0/4	0/4	4/4	0/4	4/4	0/4	4/4	0/4	4/4

To separate scale and sensor effects more precisely, we run additional (**R**) and (**RC**) seeds at 512×512 world size (32 seeds) and 128×128 world size (64 seeds), then estimate the probability of extinction and mining over simulator time with a Kaplan-Meier estimator. Extinction is rarer and delayed in the larger world, whereas mining is more frequent and occurs earlier at the smaller grid size. A Cox proportional hazards model comparing mining-discovery and extinction timing across world size and compass usage similarly indicates that larger world size substantially reduces extinction rate and enables more rapid mining discovery than smaller worlds; enabling the compass produces a comparable directional effect. Appendix B provides the full analysis.

Table 2: Mining and extinction events in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains in populations of agents with a compass (**RC**) compared to those without a compass (**R**) at the 512 × 512 grid size.  denotes the number of seeds that featured a mining event.  denotes the number of seeds in which the population went extinct before 2M steps. See Table 4 in Appendix A for detailed data from each seed.

	Island		Lake		Isthmus		Channel	
								
(RC)	4/4	0/4	2/4	2/4	4/4	0/4	3/4	1/4
(R)	3/4	1/4	2/4	4/4	1/4	3/4	0/4	4/4

We next evaluate whether the same navigational mechanism extends across terrain structure by running **Island**, **Lake**, **Isthmus**, and **Channel** worlds at 512 × 512 with (**R**) and (**RC**) agents. Table 2 reports mining and extinction counts, while Table 4 in Appendix A gives seed-level details. As in the **Beach** environment, compass sensing produces earlier and more consistent mining and lower extinction rates across these terrains. The (**RC**) populations adapt most reliably in the island and isthmus terrains, and less reliably in the lake and channel, where extinction occurs in some seeds. We hypothesize that this difference reflects terrain geometry: in the island and isthmus, moving straight east or west reliably returns an agent to water, whereas the channel and lake lack an equally simple directional rule. The (**R**) agents do not reliably adapt in any terrain, although the island condition is the least adverse baseline case. Figure 12 in Appendix D visualizes mining in the island terrain.

4.2 VISUAL FORAGING AND PREDATION

We then use the **Ocean** environment to test whether agents evolve to exploit visual sensing for resource localization and predation in an open, less directionally structured world. The comparison is between (**RCV+A**) agents, which have resource sensing, compass, a 7 × 7 visual sensor, and attacking, and (**RC+A**) agents, which have resource and compass sensors with attacking but no vision.

Figure 5 reports population, biomass utilization, move actions, and eat actions in 512 × 512 and 128 × 128 grids. Figure 9 in Appendix C provides the corresponding plots for other grid sizes, and

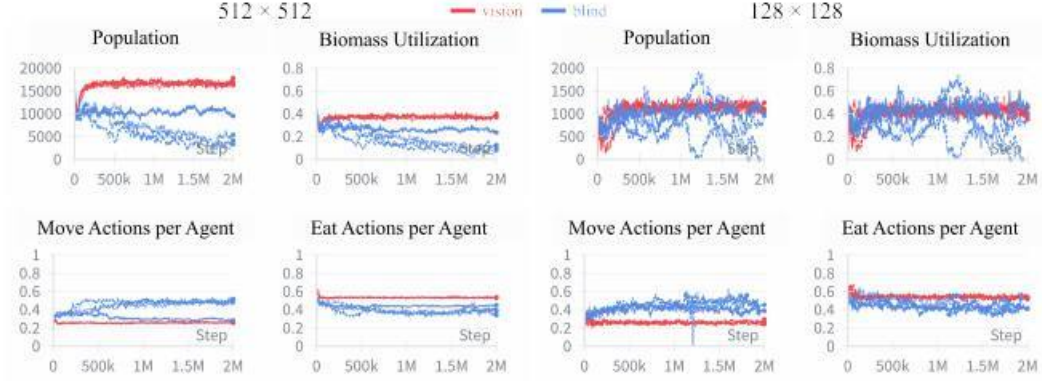


Figure 5: Population, biomass utilization, move actions per agent, and eat actions per agent in 512×512 and 128×128 **Ocean** worlds. Biomass utilization, move actions per agent, and eat actions per agent are averaged across agents at each time step. Each plot shows four seeds with vision agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). Vision produces larger and more stable populations, higher biomass utilization, less movement, and more frequent effective eating. The comparison is more stable at larger scales, indicating that larger environments make the foraging advantage of vision more consistent across seeds. Figure 9 in Appendix C compares the 1024×1024 and 256×256 grid sizes.

Table 5 in Appendix A reports summary statistics. Biomass utilization is the percentage of total environmental biomass currently carried by agents. Runs with vision (red) show less movement, more eating, larger populations, and higher biomass utilization than blind runs under the same grid sizes and attack-enabled action set. These aligned changes indicate that agents use vision to forage more efficiently rather than merely moving more often. Across environmental scales, larger worlds substantially reduce between-run variance, and the vision condition produces markedly more stable temporal dynamics.

Figure 6 reports attacks per agent and homicides per attack for the same runs, thereby separating attack frequency from attack effectiveness. Vision-equipped agents attack relatively infrequently while maintaining a successful kill rate of approximately 75%. Blind agents in large grids sometimes also achieve high kill rates, but their dynamics are substantially less stable across seeds and time. Because several factors may contribute to these high blind-agent kill rates, we do not attribute them to a single mechanism. Visual inspection (Figure 13 in Appendix D) indicates that they sometimes occur when agents cluster densely in one location, a pattern that almost never appears in the visual-agent runs. This observation is consistent with the possibility that vision also supports avoidance or spatial dispersion. At smaller scales, the visual/blind difference is less pronounced, but visual-agent populations remain more stable.

4.3 ABLATIONS

Appendix F evaluates the sensitivity of these emergent behaviors to simulator configuration and thereby clarifies which conclusions are robust to implementation-level choices. The ablations vary initial population size (F.1), resource rules (F.2), mutation rate (F.4), agent network configuration (F.5), vision range (F.6), attack strength (F.7), and agent HP (F.8). Appendix F.3 further verifies that the dynamics observed in larger environments are not reducible to self-averaging over many smaller simulations.

5 CONCLUSION

This study establishes open-ended ecological simulation as a powerful experimental setting for investigating the emergence of intelligent behavior. Across worlds that vary in scale, terrain, and sensor configuration, the experiments show that environmental extent and sensory richness materially affect whether sophisticated resource gathering, foraging, and predation behaviors emerge, and

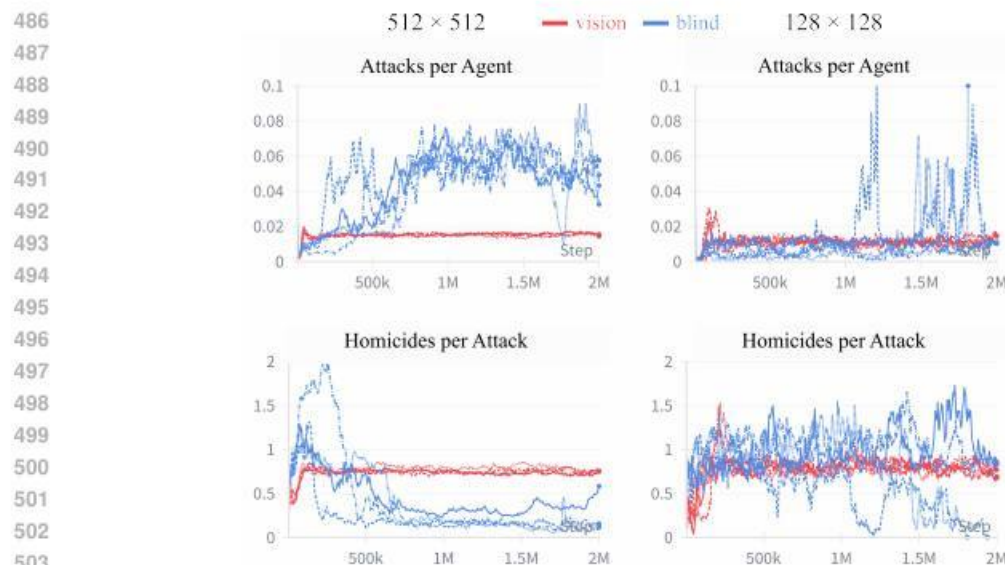


Figure 6: Attacks per agent and homicides per attack, averaged across agents, in 512×512 and 128×128 **Ocean** worlds. Each plot shows four seeds with vision agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). With vision, agents attack less often but with higher precision, consistent with the emergence of specialized predation behavior. The effect is clearer at larger scales. Figure 10 in Appendix C compares the 1024×1024 and 256×256 grid sizes.

whether those behaviors remain stable across independent runs. The results support ecological and evolutionary pressure as a viable mechanism for producing increasingly capable artificial agents in rich environments without explicit task objectives. They also demonstrate that large evolving populations can be simulated within a practical hardware budget, making ecological scale an experimentally accessible variable rather than merely a biological analogy. These findings motivate artificial ecologies as both a method for developing adaptive behavior and a scientific instrument for studying how intelligence can arise from survival, reproduction, and environmental interaction.

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A ADDITIONAL DATA

Table 3: Mining events in the **Beach** world in populations of agents with a compass (**RC**) compared to populations without a compass (**R**) across five scales.  denotes whether a run had a mining event.  denotes the percent decrease in free biomass on dry land during the mining event, and  denotes the approximate timestep at which free dry biomass reached the local maximum indicating the start of the mining event. For runs with multiple mining events, we report the first one.  denotes the timestep at which the run went extinct, or n/a if it lasted for 2M steps.

	1024x1024				512x512				256x256				128x128				64x64			
	32768				8192				2048				512				128			
Compass On (RC)																				
Seed																				
0	✓	35k	11.1	n/a	✓	105k	33.	n/a	✓	145k	49.3	n/a	✓	105k	72.5	n/a	x			55
1	✓	90k	16.2	n/a	✓	50k	17.2	n/a	✓	50k	46.6	n/a	x		140	✓	70k	85.3	675	
2	✓	40k	15.	n/a	✓	30k	14.3	n/a	✓	55k	34.7	n/a	✓	50k	50.7	n/a	✓	10k	78.	n/a
3	✓	60k	15.	n/a	✓	45k	15.6	n/a	✓	90k	47.4	n/a	✓	20k	52.	n/a	x			135
Avg	4/4	34k	14.3	0/4	4/4	58k	20.	0/4	4/4	85k	44.5	0/4	3/4	58k	58.4	1/4	2/4	40k	81.7	3/4
Compass Off (R)																				
Seed																				
0	x			n/a	x			945k	x			185k	x			65k	x			45k
1	✓	275k	12.4	n/a	x			1085k	x			110k	x			55k	x			40
2	x			n/a	x			1070k	x			165k	x			55k	x			25
3	x			n/a	x			280k	x			125k	x			90k	x			40
Avg	1/4	275k	12.4	0/4	0/4			n/a	0/4			4/4	0/4			4/4	0/4			4/4

Table 4: Mining events in the **Island**, **Lake**, **Isthmus**, and **Channel** among (RC) agents compared to (R) agents. Grid size is 512×512 with an initial population of 8192.

	Island				Lake				Isthmus				Channel			
Compass On (RC)																
Seed																
0	✓	40k	94.3	n/a	x			45k	✓	45k	33.	n/a	✓	50k	27.4	n/a
1	✓	30k	94.3	n/a	✓	30k	22.5	n/a	✓	55k	37.3	n/a	✓	80k	30.8	n/a
2	✓	30k	94.8	n/a	x			45k	✓	40k	28.1	n/a	✓	50k	41.	n/a
3	✓	70k	95.1	n/a	✓	35k	25.3	n/a	✓	30k	41.5	n/a	x			80k
Avg	4/4	43k	94.6	0/4	2/4	33k	23.9	2/4	4/4	43k	35.	0/4	3/4	60k	33.1	1/4
Compass Off (R)																
Seed																
0	x			185k	x			125k	x			145k	x			100k
1	✓	75k	30.9	n/a	x			130k	x			260k	x			105k
2	✓	85k	47.	n/a	✓	75k	23.4	1400k	✓	95k	18.8	n/a	x			105k
3	✓	115k	39.5	n/a	✓	70k	24.3	1250k	x			140k	x			100k
Avg	3/4	92k	39.1	1/4	2/4	73k	23.9	4/4	1/4	95k	18.8	3/4	0/4			4/4

Table 5: Population and foraging data for Vision and Blind agents in the **Ocean** world at different scales. denotes population size. denotes biomass utilization, the percentage of total biomass in the system that is currently inside the agents' stomachs. denotes move actions per agent, the fraction of time an agent took a move action, averaged across agents. Similarly, denotes the proportion of agents' actions that took an eat action. Reported values of , , , and are averages of the metric over all steps for the seed.

	1024x1024				512x512				256x256				128x128			
	32768				8192				2048				512			
Vision (RCV+A)																
Seed																
0	58354	.396	.402	.426	16413	.369	.257	.532	4195	.382	.255	.536	1126	.426	.261	.541
1	65107	.365	.257	.529	16200	.364	.256	.533	4142	.377	.258	.536	1103	.415	.262	.537
2	64735	.363	.257	.530	16560	.372	.257	.533	4269	.388	.260	.536	1041	.392	.254	.542
3	64595	.362	.256	.532	16206	.365	.255	.531	4249	.388	.259	.533	1100	.414	.257	.538
Avg	63198	.371	.293	.504	16345	.368	.256	.532	4213	.384	.258	.535	1092	.412	.259	.539
Blind (RC+A)																
Seed																
0	22734	.143	.446	.372	6549	.173	.442	.400	2036	.209	.433	.461	545	.219	.465	.427
1	23670	.151	.426	.388	6939	.180	.438	.396	3107	.321	.390	.416	1082	.444	.416	.460
2	24745	.159	.358	.426	5448	.136	.474	.374	1587	.163	.440	.433	979	.421	.404	.436
3	20004	.130	.469	.359	10354	.265	.311	.439	3164	.324	.321	.450	788	.317	.431	.477
Avg	22788	.146	.425	.386	7323	.189	.416	.402	2473	.254	.396	.440	848	.351	.429	.450

Table 6: Predation data for Vision and Blind agents in the **Ocean** world at different scales.  denotes attacks per agent, the fraction of time an agent took the attack action, averaged across all agents. The  denotes homicides per attack, the number of agents killed per attack action. Reported values of  and  are averages over all steps for the seed. Note the outlier of 0.562 for  of seed 3 in the 128×128 (**RC+A**) run. This seed went extinct before 2M steps, resulting in a spike in the attacks per agent metric due to dwindling population size.

	1024x1024		512x512		256x256		128x128	
	32768		8192		2048		512	
Vision (RCV+A)								
Seed								
0	.011	.776	.015	.731	.013	.784	.010	.788
1	.016	.733	.015	.741	.014	.780	.011	.793
2	.016	.707	.015	.737	.014	.750	.012	.771
3	.016	.712	.015	.780	.013	.759	.011	.765
Avg	.015	.732	.015	.747	.014	.758	.011	.779
Blind (RC+A)								
Seed								
0	.070	.218	.042	.437	.005	1.074	.016	.558
1	.054	.300	.044	.383	.016	.871	.006	1.032
2	.076	.236	.048	.238	.031	.383	.010	.979
3	.059	.203	.044	.456	.033	.446	.562	.841
Avg	.065	.239	.045	.378	.021	.693	.148	.853

B DEMOGRAPHY

We compare rates of extinction and mining emergence dependent on world size and agent access to global direction via the compass sensor. Extinction is defined as the population reaching zero individuals, while mining behavior is defined as it was in earlier experiments (a 10% drop in free biomass on land).

Runs are grouped based on both world size and compass access. We simulated 64 runs each of 128×128 with no compass and 128×128 with compass, and 32 runs each of 512×512 with no compass and 512×512 with compass. For each of the four groups, we fit a non-parametric Kaplan–Meier estimator to the time-to-extinction distribution (treating runs that did not go extinct within 2M steps as right-censored). In Figures 7 and 8 we observe that worlds tend to survive longer and produce mining more quickly if they are large or if agents have compass information.

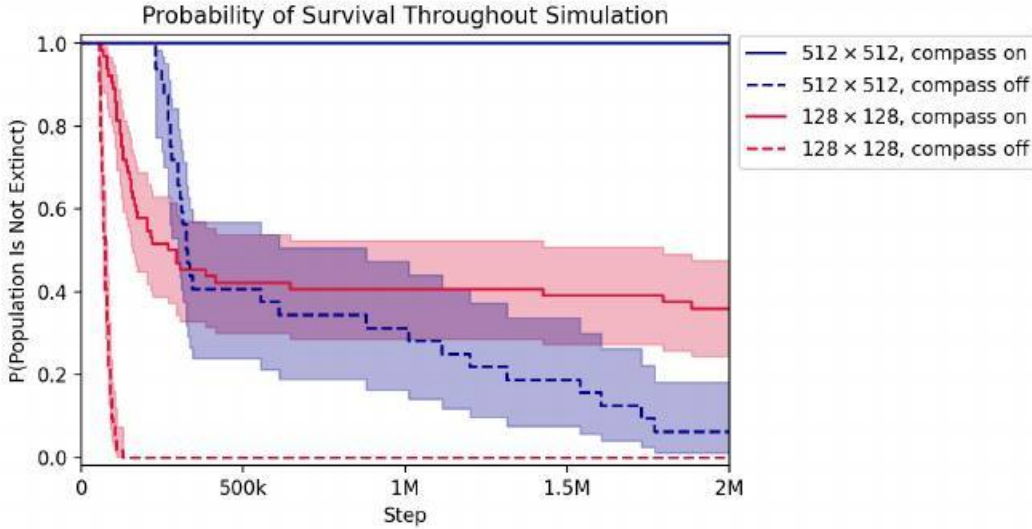


Figure 7: Probability the population has not gone extinct given a world size, compass access, and timestep. We fit a unique non-parametric Kaplan–Meier estimator for each group of runs (e.g. 512×512 , compass on). The shaded areas represent the 95% confidence intervals for each curve. Each 512×512 group fits the estimator on 32 runs; each 128×128 group fits on 64 runs.

We compare emergence and extinction timing with a Cox proportional hazards model, a semi-parametric survival model relating covariates to event hazard rates (Table 7). Runs that did not experience an emergence or extinction event by the end of 2M time steps were treated as censored. For extinction, both larger world size and compass use substantially reduce the hazard; the extinction rate in 128×128 is about 25 times greater than the corresponding rate in 512×512 worlds ($HR=0.04$). Enabling the compass produces an effect of a similar magnitude ($HR=0.03$). For mining emergence, larger worlds generate mining behavior much more rapidly ($HR=6.07$), and compass use accelerates emergence dramatically ($HR=34.59$). The models show good discriminative ability between groups (concordance 0.80 for extinction, 0.85 for mining). We find that larger worlds can sustain larger populations, reducing the chance of hitting the absorbing zero boundary while simultaneously exploring a larger phenotype space due to greater quantities of mutation as compared to smaller worlds.

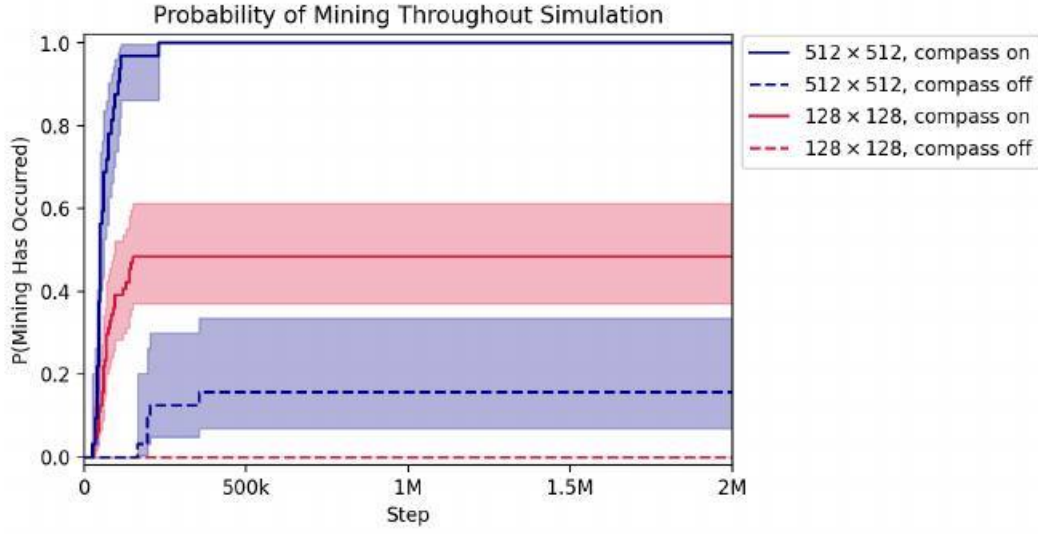


Figure 8: Probability mining has evolved within the population given a world size, compass access, and timestep. We fit a unique non-parametric Kaplan-Meier estimator for each group of runs (e.g. 512×512 , compass on). The shaded areas represent the 95% confidence intervals for each curve. Each 512×512 group fits the estimator on 32 runs; each 128×128 group fits on 64 runs.

Table 7: Cox proportional hazards models for extinction and emergence events with a 128×128 , compass off baseline. Hazard ratios (HR) less than 1 indicate reduced hazard (longer time to event); for example, extinction $HR = 0.04$ for 512×512 means extinction occurs at 0.04 times the rate of 128×128 . There are 192 total runs: 32 seeds with compass on and 32 seeds with compass off in the 512×512 world, and 64 seeds with compass on and 64 seeds with compass off in the 128×128 world.

Covariate	Extinction			Mining		
	HR	95% CI	p-value	HR	95% CI	p-value
Size 512×512	0.04	0.02–0.07	< 0.005	6.07	3.61–10.20	< 0.005
Compass On	0.03	0.01–0.05	< 0.005	34.59	13.51–88.57	< 0.005
Concordance	0.80			0.85		
Events / Total	135 / 192			68 / 192		

C ADDITIONAL FORAGING AND PREDATION RESULTS

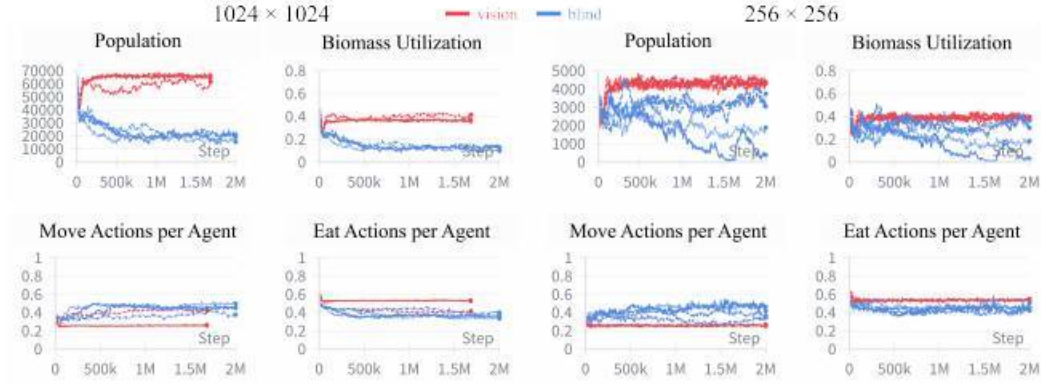


Figure 9: Foraging behaviors in the **Ocean** world at different scales. Enabling vision in agents (red) results in more stable populations and greater biomass utilization as a result of more frequent and effective eating and less movement. At larger scales, these dynamics are more stable with less variance across runs. Note that due to limitations on GPU hours, the vision seeds at the 1024×1024 scale ran for 1.7M instead of 2M steps.

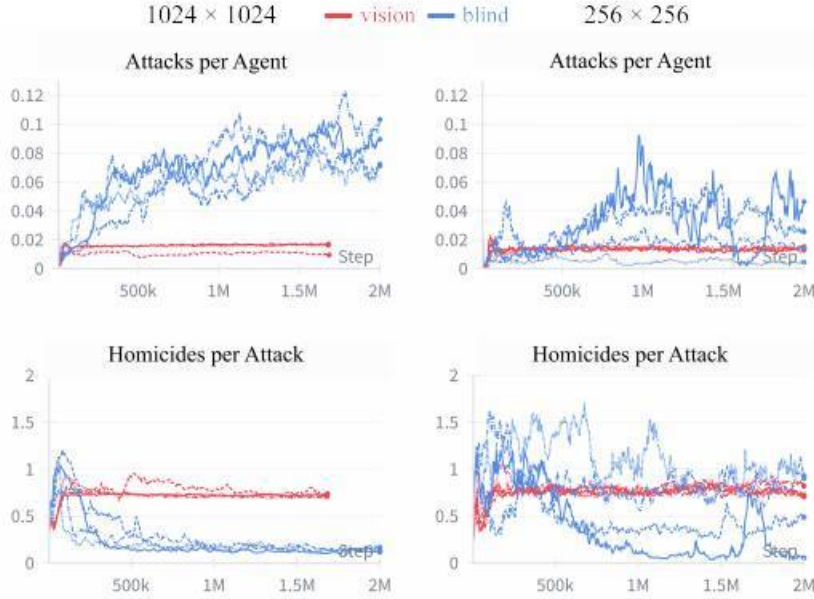


Figure 10: Predation behaviors in the **Ocean** world at different scales. With vision (red), agents attack less frequently but with much higher precision, suggesting the emergence of specialized predation behaviors in addition to foraging capabilities (Figure 9). This dynamic is again more pronounced at larger scales, with more variance in the behaviors of blind agents (blue) in smaller world sizes.

D ADDITIONAL EXPERIMENT VISUALIZATIONS

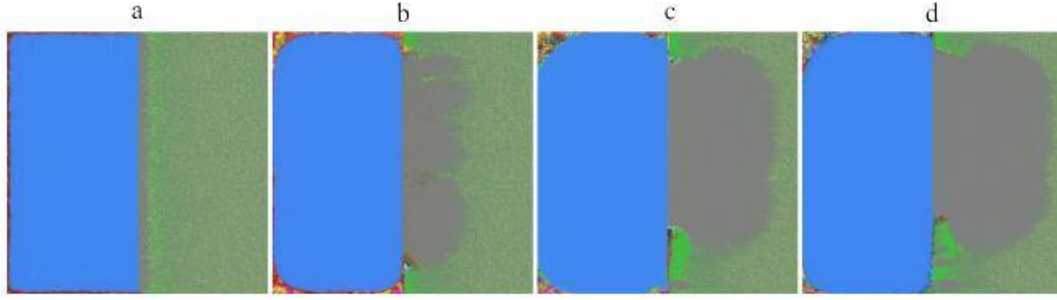


Figure 11: Larger scale environments induce more varied and complex ecologies that give rise to unique visual patterns. In this 1024×1024 **Beach** world, (RC) agents initially died at the shore (a), transferring biomass to dry land. They then developed mining behavior, clearing biomass from the beach (b). When biomass re-accumulated on the ends of the beach (c), subpopulations of agents re-developed mining behavior, traveling inland on these mini biomass beaches to extract the resources (d).

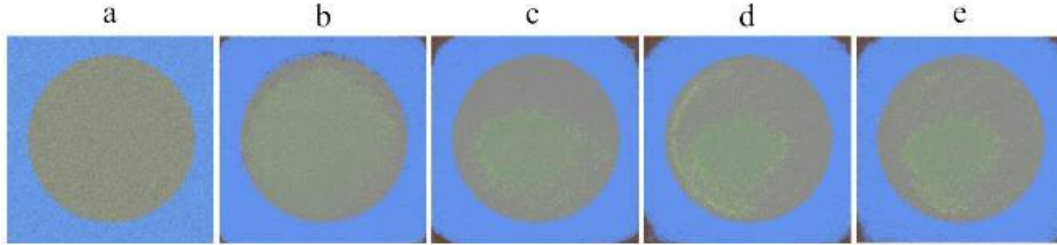


Figure 12: Mining behavior emerges in varied terrains. In this 512×512 **Island** world with (RC) agents, different regions of the shore are mined throughout the run (b) (c). Biomass is transferred back to parts of the island once mining becomes difficult and agents die before they can return to water (d). Agents adapt to re-mine this wall of accumulated biomass, clearing it from shore (e).

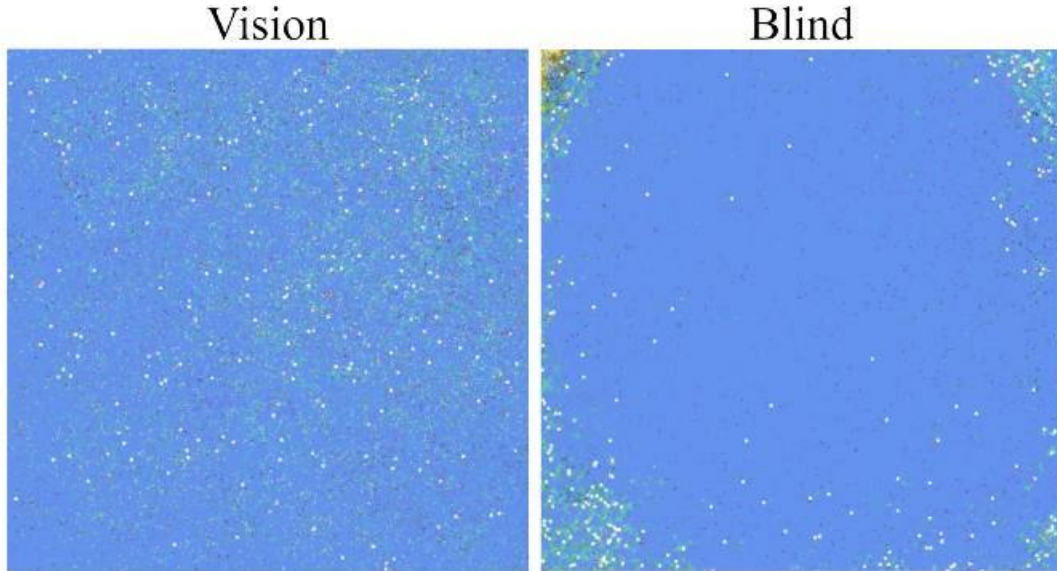


Figure 13: Representative runs in a 512×512 **Ocean** world where agents have attacking enabled: vision (RCV+A) agents on the left and blind (RC+A) agents on the right. The white squares indicate an agent is attacking at that location. With vision, agents adapt to locate free resources throughout the ocean world and prey on other agents more effectively (see Figures 5 and 6). Blind agents explore the ocean for resources to a much lesser extent, instead adapting policies for clustering in the corners of the world.

E DETAILS ON COMPUTATION AND REPRODUCIBILITY

We report details on simulation design, runtime, and experiment reproducibility.

Simulator design. We conduct all experiments using a JAX-based, GPU-accelerated simulator designed for large-scale multi-agent ecosystems with open-ended populations. Our simulator computes every part of the environment step on the GPU, including terrain dynamics, resource flows, agent updates, and measurement, so that policy evaluation and simulation share a single JAX compilation pipeline. The simulator exposes a minimal population-level interface: given current map state, agent states, and heritable traits, a population algorithm supplies actions and updated traits and handles births, deaths, and environmental transitions. This separation allows the same environment implementation to be reused with different evolutionary algorithms, reinforcement learning methods, or other population update rules. All operations are implemented as pure JAX functions combined with vectorization and just-in-time compilation, enabling fusion into a small number of GPU kernels. This yields the throughput required for our scaling experiments, with runtime measurements for a single H100 GPU given in Table 8.

A key feature of the simulator is its built-in tooling, including a measurement API for computing and recording aggregate statistics such as biomass utilization, movement and eating rates, attack and homicide rates, and extensive environment-level parameters at each step, which we use directly in our figures and tables. In addition, an interactive 3D viewer renders terrain, water, resources, and agents from generated reports, and supports inspection of individual agents (Figure 1). These tools were critical for discovering and validating qualitative phenomena such as long-range resource mining and predation patterns that are not obvious from scalar metrics alone.

From a high-performance computing perspective, our simulator fills a gap relative to existing simulators. General-purpose agent-based frameworks such as MASON (Luke et al. 2005), FLAME-GPU (Richmond et al. 2024), or Griddly (Bamford et al. 2020) offer high configurability but require substantial environment engineering and do not directly target the open-ended, evolving populations at the scales we study. Conversely, GPU-native multi-agent RL environments emphasize fixed numbers of agents and task-specific reward structures. We combine GPU-native execution, explicit support for births and deaths with mutable traits, and configurable resource and terrain dynamics, enabling open-ended populations of over 10^5 agents while remaining interoperable with diverse population algorithms. This combination yields a state-of-the-art platform for scaling ecological experiments like those in this paper and a useful foundation for future work on emergent behavior in large populations. Our simulator will be released upon acceptance.

Runtime data. Table 8 shows runtime data across different grid sizes and maximum populations of agents. Each run was conducted on a single H100 GPU for a maximum of 2M steps. Note that while this table shows run-times for (RC) and (RCV + A) agents with 2 MLP layers and 64 channels per layer ($\sim 22k$ parameters total), the runtime varies with the size of the agent networks, the available sensors, and the level of logging.

Our largest runs at the 1024×1024 grid size take ~ 10 hours in the **Beach** terrain and ~ 14 hours in the **Ocean** terrain on one H100, with memory usage of roughly 60GB. **Ocean** runs are slightly slower due to the enabled agent attack action and vision sensing.

Table 8: Runtime data in the **Beach** and **Ocean** worlds across different grid sizes and maximum populations of agents. The (RC) agents each have a 22,465 parameter neural network. The (RCV + A) agents each have a 22,529 parameter neural network. Simulation steps per second (hz) and total extrapolated wall-clock runtime in hours for 2M steps are reported. Effective runtime is less for seeds in which populations go extinct before 2M steps.

Environment	1024×1024		512×512		256×256		128×128		64×64	
	hz	runtime	hz	runtime	hz	runtime	hz	runtime	hz	runtime
Beach (RC)	55	10.1 hrs	190	2.9 hrs	315	1.8 hrs	535	1.0 hrs	750	0.7 hrs
Ocean (RCV + A)	40	13.9 hrs	142	3.9 hrs	235	2.4 hrs	415	1.3 hrs	750	0.7 hrs

Reproducibility. Due to the large and randomized nature of our environments, there can be substantial differences in the precise outcomes of each experiment from run to run. As a result, we report the results of four different random seeds for each experiment. Unfortunately, due to nondeterminism in some low-level GPU operations, we find that repeating an experiment with the same random seed does not reliably produce the same results. In particular, modern GPUs may execute parallel floating-point operations such as summations in different orders across runs and fuse operations in ways that change rounding behavior. Very small differences arising from finite-precision arithmetic can accumulate over many steps into noticeable drift in the simulation trajectory. The use of 16-bit weights can coarsen these rounding effects, but is not the primary source. Despite differences across runs, clear high-level patterns emerge when the results of multiple seeds are taken in aggregate.

F ABLATIONS

We present a series of ablations that evaluate how specific hyperparameters and environment or agent configurations affect the robustness of the main emergent-behavior results.

F.1 INITIAL POPULATION SIZE SWEEP FOR FIXED MAP SIZE

We evaluate how initial population size affects the consistency of mining behavior and population stability in a 256×256 **Beach** world. The reported results are for **(RC)** agent populations. Agents without compass access (**R**) had 0% mining rate and 100% extinction rate across all initial population sizes except 8192, the largest value tested, where they had 25% mining and 75% extinction frequency. Mining and extinction frequencies need not sum to 100%: a population may develop mining and later go extinct, or it may persist for the full 2M experimental steps without developing mining behavior.

Figure 14 shows that initial population size has no significant effect on mining consistency or extinction rates in this setting: populations initialized with only 128 agents evolve mining as frequently as populations initialized with 8192 agents, and extinction rates remain largely stable across the sweep.

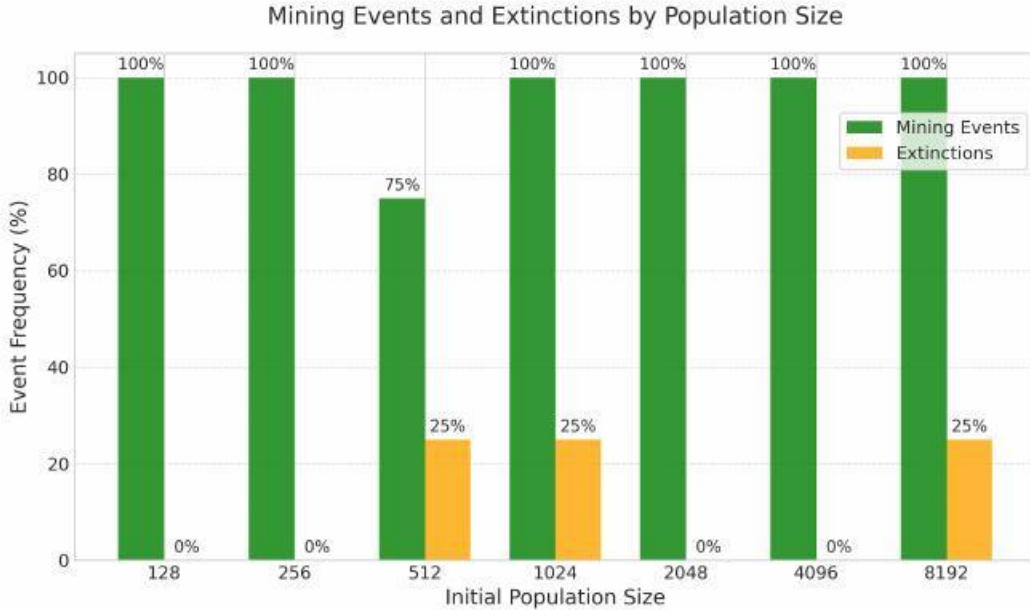


Figure 14: Frequency of mining and extinction events in a 256×256 **Beach** world for different initial sizes of **(RC)** agent populations. Frequencies are computed across four random seeds for each initial population size; i.e. 25% Extinction Event Frequency means that in 1/4 seeds, the population went to 0 before 2M simulation steps.

Figure 15 explains why the initial population has little visible effect. Regardless of initialization, each run rapidly approaches a similar population size within the first 50,000 simulation steps. This pattern indicates that the environment imposes an implicit carrying capacity through the fixed resources available in the system: populations initialized below that capacity grow toward it, whereas populations initialized above it are constrained by resource scarcity. Consequently, effective population size becomes similar after an early correction phase, leaving mining and extinction frequencies largely unchanged. Larger initial populations nevertheless contribute modestly to total system resources because agents carry biomass that is deposited upon death; accordingly, they reach higher maximum populations on average over the course of simulation (Table 9), although not enough to alter mining or extinction frequency in this setting. Early convergence also does not imply long-run identity across seeds: later population sizes can diverge substantially as agents evolve different abilities at different times and thereby reshape subsequent resource distributions.

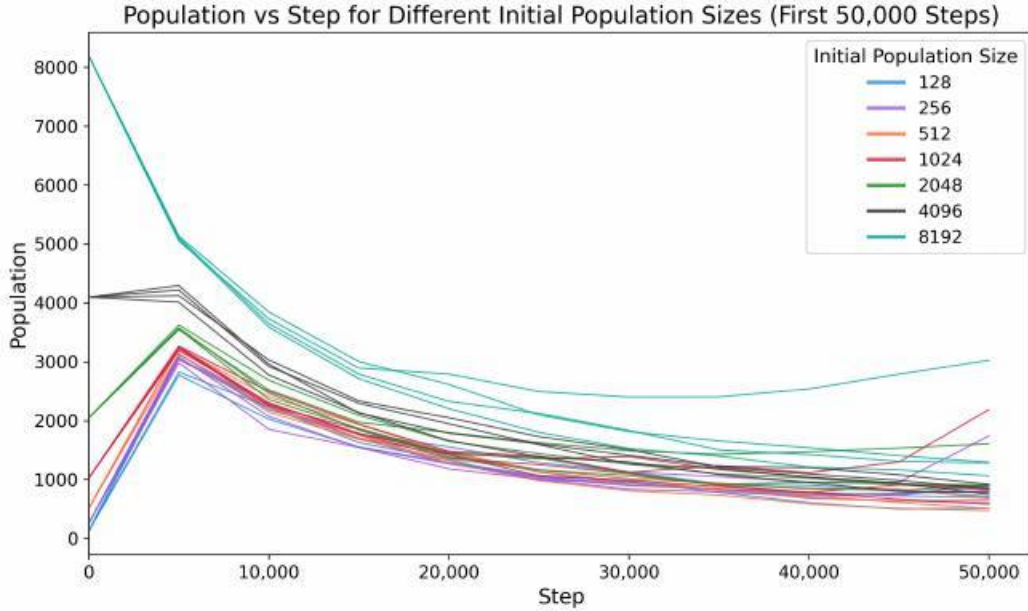


Figure 15: Population size during the first 50,000 time steps in the 256×256 **Beach** with **(RC)** agents, with four seeds per initial population size. The environment supports an implicit carrying capacity based on resources available in the system, and all initial population sizes approach this value early in the run. Larger initial population sizes contribute to the total resources in the system due to biomass deposited on agent death, raising the carrying capacity somewhat.

F.2 VARYING RESOURCE RULES

We evaluate how two resource hyperparameters, biomass photosynthesis rate and initial biomass site density, affect mining and extinction frequency. The setting matches Appendix F.1: the initial population size is fixed at 2048, and agents again have compass access **(RC)**.

The biomass photosynthesis rate controls the rate at which landscape biomass creates energy through photosynthesis, thereby replenishing resources over time. The initial biomass site density determines the fraction of grid cells containing biomass at the start of simulation. All experiments in the main body (Section 4) use a biomass photosynthesis rate of 0.01 and an initial biomass site density of 1/4.

Whereas varying initial population size does not affect mining and extinction consistency (Figure 14), changing resource rules has a significant effect, as shown in Figures 16 and 17.

Larger biomass photosynthesis rates and higher initial biomass site densities directly increase the effective carrying capacity of the environment by increasing the resources available in the system. Larger populations can therefore be sustained throughout the run, increasing the likelihood that mutations discover mining behavior while lowering the probability of extinction. The data support this

Table 9: The maximum population size reached after step 0 for different starting population sizes, with averages and standard deviations across four random seeds. Larger initial population sizes reach slightly higher maximum populations with greater variance across seeds than for lower initial population sizes. However, the max population size reached is largely robust to the initial population size: an initial population size (128) of only 1.6% of an initial size of 8192 reaches 78.9% of the max population reached by the latter (3178 vs. 4026).

Initial Population Size	Max Population Size After Step 0
128	3178 $\pm$ 217
256	3056 $\pm$ 61
512	3451 $\pm$ 437
1024	3613 $\pm$ 381
2048	3786 $\pm$ 431
4096	3689 $\pm$ 437
8192	4026 $\pm$ 642

interpretation: omitting seeds that went extinct, runs with initial biomass site density 1/16 reached an average maximum population of 1608 after step 0, compared with 2466 for density 1/8, 3786 for density 1/4, and 5353 for density 1/2.

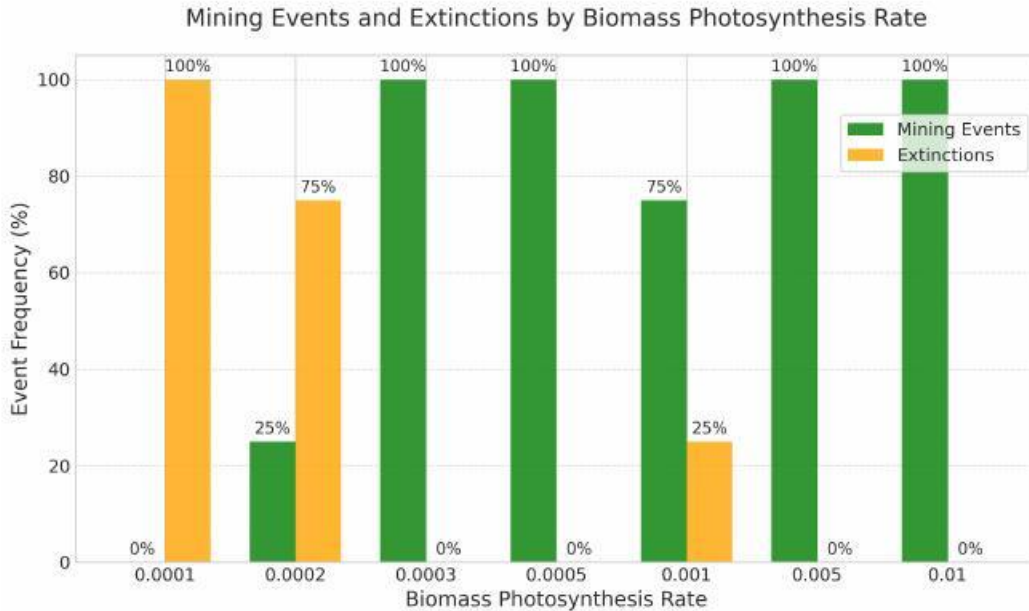


Figure 16: Frequency of mining and extinction events in a 256×256 **Beach** world for different settings of the biomass photosynthesis rate resource parameter. Frequencies are computed across four random seeds for each setting. Mining consistently emerges with biomass photosynthesis rates of 0.0003 and greater.

F.3 RULING OUT SELF-AVERAGING IN LARGER POPULATIONS

A possible alternative explanation for the increased stability of mining behavior in large environments (Table I) is self-averaging rather than qualitatively different dynamics. By self-averaging, we mean the statistical-physics intuition that, as system size grows, fluctuations in macroscopic quantities shrink roughly as $1/\sqrt{N}$, where N is the system size, so that a single large system behaves like the average of smaller systems with the same total size. Under this view, a 512×512 **Beach** world might simply act like many loosely coupled 128×128 worlds run in parallel. If so, the scaling

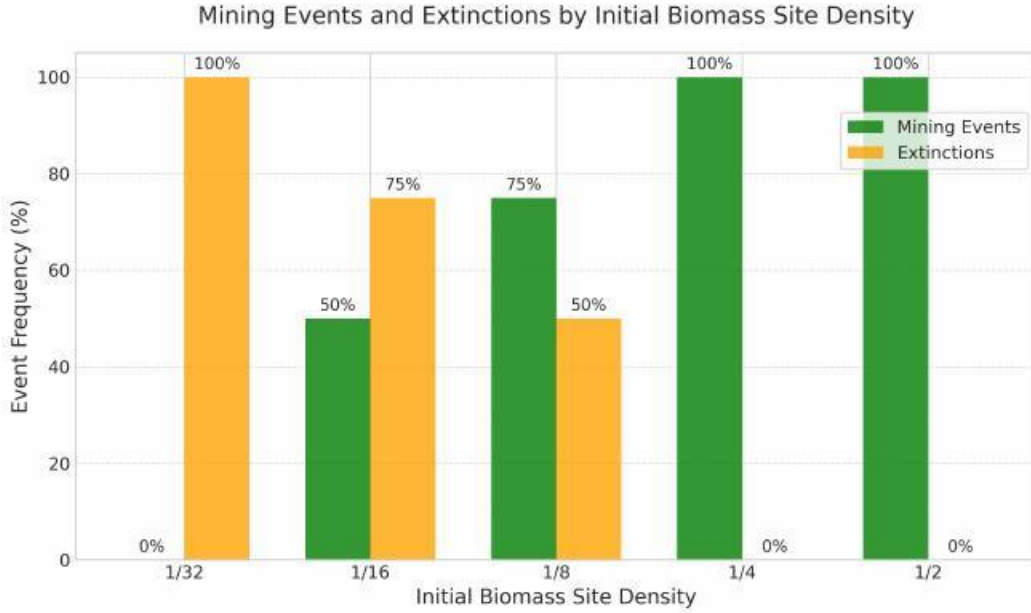


Figure 17: Frequency of mining and extinction events in a 256×256 **Beach** world for different settings of the initial biomass site density resource parameter. Frequencies are computed across four random seeds for each setting. Greater initial biomass site densities result in more consistent discovery of mining and lower chance of extinction.

results could in principle be reproduced by running enough small worlds and averaging over random seeds.

To test this hypothesis, we hold the total “sample size” approximately fixed while changing how it is partitioned across environments. In the **Beach** terrain with compass-equipped agents (**RC**), we consider three conditions:

- 4 runs of a 512×512 world with 8192 initial agents each,
- 16 runs of a 256×256 world with 2048 initial agents each,
- 64 runs of a 128×128 world with 512 initial agents each.

In each case, the total number of grid cells across runs, the total starting resources and the total initial number of initial agents are identical:

$$4 \times 512^2 = 16 \times 256^2 = 64 \times 128^2, \quad 4 \times 8192 = 16 \times 2048 = 64 \times 512.$$

Each seed is run for 2M steps with the same hyperparameters as in Section 4.1. As a control, we repeat the protocol for agents without a compass (**R**). Table 10 summarizes the frequency of long-distance mining and extinction in each configuration.

If large environments acted as self-averages of many smaller ones, mining and extinction rates should be roughly invariant to how grid cells and agents are distributed across runs: 4 large worlds and 64 small worlds would exhibit similar probabilities of discovering mining. Instead, the outcomes depend strongly on world size. For (**RC**) agents, mining emerges in all runs with no extinctions at 512×512 , remains common but not guaranteed at 256×256 , and becomes unreliable and fragile at 128×128 , where many populations go extinct before reaching 2M steps. In contrast, (**R**) agents never develop mining and go extinct at all scales, confirming that the compass remains a key enabler of long-range behavior.

These results rule out the simple self-averaging explanation tested here. Larger environments enable and stabilize mining behavior in ways that cannot be explained solely by averaging over more small simulations.

Table 10: Mining and extinction events when trading off world size against the number of independent runs. We compare **Beach** worlds at grid sizes 512×512 , 256×256 , 128×128 with initial populations scaled proportionally (8192, 2048, 512 agents, respectively). For each size, we allocate the same total number of grid cells and initial agents across all runs by running 4 seeds at 512×512 , 16 seeds at 256×256 , and 64 seeds at 128×128 . We report the fraction of runs in which mining emerged () and the fraction in which the population went extinct before 2M steps () for agents with a compass (**(RC)**) and without a compass (**(R)**).

	512×512		256×256		128×128	
	8192		2048		512	
						
(RC)	4/4	0/4	15/16	1/16	31/64	41/64
(R)	0/4	4/4	0/16	16/16	0/64	64/64

Even when the total number of cells, agents, and simulation steps is matched across conditions, concentrating resources into a few large worlds produces a regime in which mining is consistent and extinctions are rare, whereas dispersing the same resources across many small worlds yields populations that frequently fail to discover mining and are more likely to collapse. This pattern indicates that increasing environmental scale is not merely smoothing stochastic noise; it is altering the underlying ecological dynamics, for example by supporting larger, more spatially distributed populations and more persistent diversity in evolving policies.

F.4 MUTATION RATE SWEEP

We test the sensitivity of the emergence of mining to the mutation rate hyperparameter, which controls the scale of Gaussian noise added to a child agent’s policy upon birth. Intuitively, the mutation rate directly impacts how different a child agent is from its parent. We once again run experiments in a 256×256 **Beach** world with an initial population of 2048 (**(RC)**) agents for a maximum of 2M steps.

In Figure 18 we show that between mutation rates of 6×10^{-3} and 7.875×10^{-2} , mining emerges in at least 3 of the 4 seeds run, indicating that our results in Section 4.1 with mutation rate 3×10^{-2} are robust within this interval. As expected, mutation rates that are too small (e.g., 3×10^{-3}) prevent agents from developing new behavior and impair survival, leading to extinction. Conversely, large mutation rates (e.g., 3×10^{-1}) undermine population stability, consistently leading to extinction with no mining adaptation.

F.5 NETWORK SIZE

We study the effect of the size and architecture of the agent policy network on mining and extinction frequency, as well as the impact of environmental scale on this effect. We investigate 256×256 and 512×512 **Beach** worlds with initial population sizes of 2048 and 8192 (**(RC)**) agents, respectively (scaling by four with the number of grid cells). We ran baselines with (**(R)**) agents for each experiment, again finding rare to no mining events and 100% extinction rates.

The main experiments in Section 4 used agent policies with 2 layers and 64 channels, yielding a network of 22465 parameters per agent (Table 11). In Figure 19 we show that in the 256×256 **Beach**, halving channels to 32 (resulting in 9185 total parameters) or doubling to 128 (giving 61313 parameters) with number of layers fixed has minimal to no effect on mining and extinction frequency. Meanwhile, changing network depth has a significant effect: all populations of agents with just 1 layer and 32 channels went extinct, and 50% of populations of agents with 3 layers and 128 channels went extinct.

Crucially, scaling the environment to the 512×512 grid size produced significantly different dynamics for networks of varying depth (Figure 20). In particular, in the larger world, agent populations with 3 layer, 128 channel networks consistently mined and survived through 2M steps. Surprisingly, agent populations with the small 1 layer, 32 channel networks mined consistently as well. We hy-

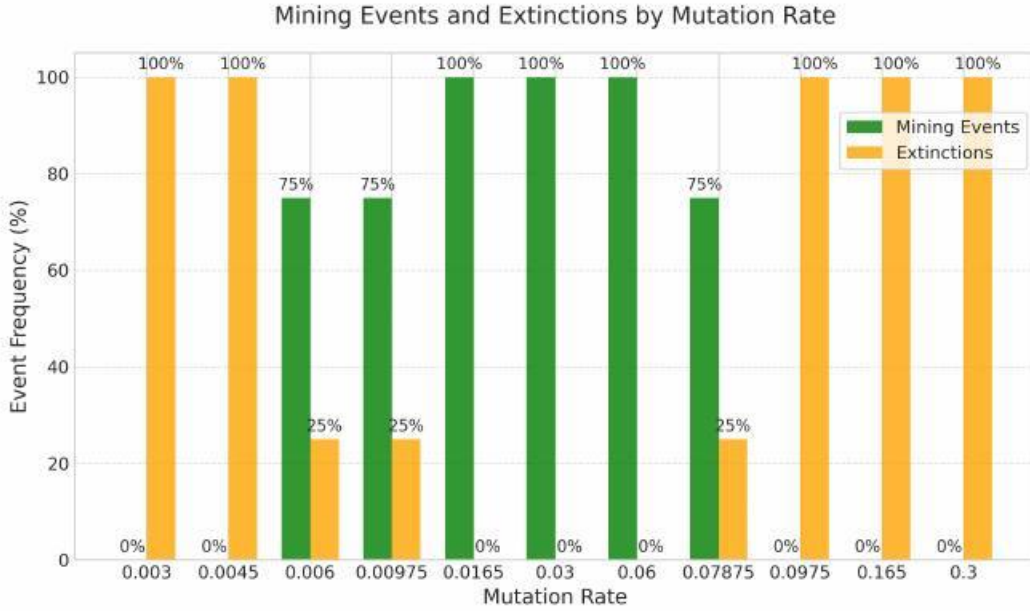


Figure 18: Frequency of mining and extinction events in a 256×256 **Beach** world for different mutation rates. Frequencies are computed across four random seeds for each setting.

pothesize that it is more challenging for both smaller and larger networks to discover new behaviors for different reasons. When the network is smaller, the reduced capacity makes it hard to find good solutions. When the network is larger, the increased size of the search space means it takes longer to find promising new behavior. Regardless of the reasons, the fact that these networks fail so often in small environments but succeed in larger ones further illustrates the stability that comes from increasing scale. The larger environment can support larger populations, which allows for more exploration in policy space. Simultaneously, it takes longer for locally successful mutants to propagate across a large environment, which promotes diversity and again ensures greater exploration. For more discussion of these effects see (Mitchell et al., 2006).

Table 11: Total parameters in (RC) and (R) agent policies with different network architectures.

Architecture (layers, channels)	Parameter Count	
	(RC)	(R)
1, 32	8129	7969
2, 32	9185	9025
2, 64	22465	22145
2, 128	61313	60673
3, 128	77825	77185

F.6 VARYING VISION RANGE

We assess the impact of different vision ranges and viewing configurations on predation and foraging behaviors in the 512×512 **Ocean** world with (RCV+A) agents. We evaluate visual sensor ranges of 3×3 , 7×7 , and 9×9 . For each range, we evaluate two viewing configurations shown in Figure 21 a) Around: agents can see grid cells all around them b) Front: agents cannot see cells behind them. The total number of cells in the agent’s vision range is equal for the two configurations. The main experiments with vision in Section 4.2 were run with the 7×7 Around setting.

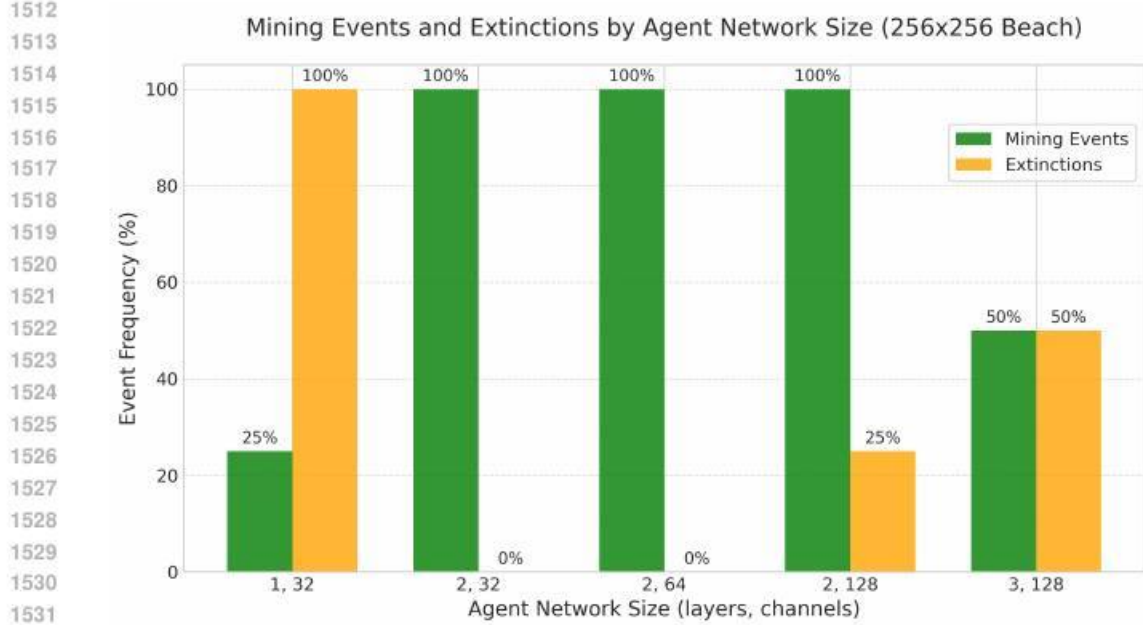


Figure 19: Frequency of mining and extinction events in a 256×256 **Beach** world for different agent policy network sizes. Frequencies are computed across four random seeds for each setting.

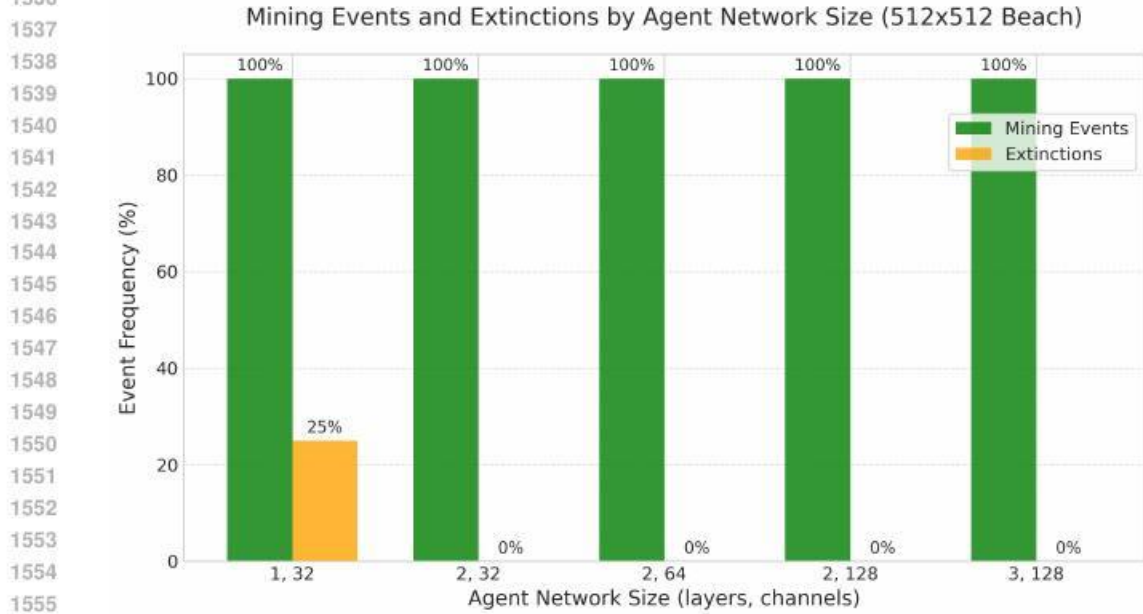


Figure 20: Frequency of mining and extinction events in a 512×512 **Beach** world for different agent policy network sizes. Frequencies are computed across four random seeds for each setting.

These results indicate that a smaller 3×3 vision range results in behavior that is largely similar to no vision at all (Blind), while the results at 9×9 appear similar to 7×7 . This indicates that there may be diminishing returns of larger vision ranges in the current environmental configuration. This is intuitive given the rest of the environmental configuration; since the agents have no memory, it is unlikely that visual features at the periphery of a 9×9 window could meaningfully alter the optimal action relative to a 7×7 window consistently enough to provide a strong learning signal.

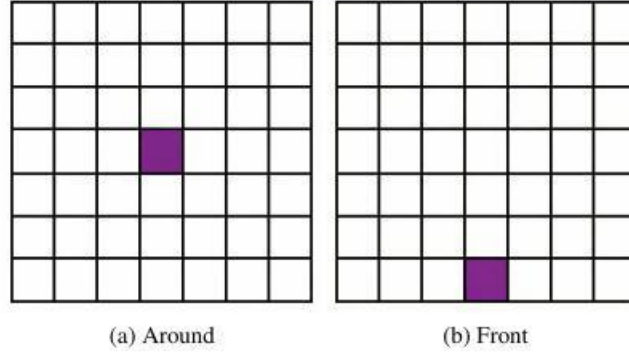


Figure 21: Comparison of the Around and Front viewing configurations for the 7×7 vision range. The purple cell represents an agent, facing the top of the page.

Table 12: Foraging and predation data in the 512×512 **Ocean** world for vision agents (**RCV+A**) with varying vision ranges and viewing configurations. Refer to Tables 5 and 6 for definitions of the metrics represented by each icon. For each seed, the value for each metric was averaged across all steps starting at step 500k to allow them to stabilize. Reported values are the mean across the four seeds and the standard deviation of the four seed averages.

Vision Range	Viewing Configuration						
3×3	Blind	7212 ± 2496	$.181 \pm .060$	$.435 \pm .090$	$.392 \pm .033$	$.032 \pm .006$	$.307 \pm .090$
	Around	10308 ± 4016	$.258 \pm .105$	$.352 \pm .014$	$.427 \pm .022$	$.040 \pm .030$	$.388 \pm .195$
	Front	13863 ± 520	$.351 \pm .023$	$.336 \pm .008$	$.437 \pm .009$	$.026 \pm .006$	$.470 \pm .145$
7×7	Around	16739 ± 122	$.373 \pm .003$	$.259 \pm .000$	$.532 \pm .001$	$.015 \pm .000$	$.751 \pm .007$
	Front	15255 ± 623	$.343 \pm .015$	$.253 \pm .001$	$.540 \pm .004$	$.015 \pm .000$	$.690 \pm .041$
9×9	Around	16744 ± 117	$.372 \pm .002$	$.257 \pm .001$	$.534 \pm .001$	$.015 \pm .000$	$.728 \pm .009$
	Front	15211 ± 1416	$.342 \pm .033$	$.253 \pm .001$	$.541 \pm .007$	$.015 \pm .000$	$.664 \pm .050$

The data also suggest that at the 7×7 and 9×9 vision ranges, the Around configuration is slightly more favorable in terms of population stability and effectiveness of attack and eat actions, resulting in higher average population , biomass utilization , and homicides per attack  than the Front configuration. We hypothesize that this is due to diminishing returns of being able to see more distant pixels; in the Around configuration, more pixels in the agent’s vision range are nearby, likely containing more relevant information for near-term actions than distant cells visible with the Front configuration. At the 3×3 vision range, however, the trend is reversed, with the Front configuration being more favorable than Around; this indicates that in this environment, the marginal benefit of seeing one more pixel ahead through the 3×3 Front configuration outweighs that of seeing one pixel behind in the 3×3 Around configuration, which gives most similar behavior to the Blind setting.

F.7 VARYING ATTACK STRENGTH

We study the effect of reducing agent attack strength on emergent behaviors and population dynamics in the 256×256 **Ocean** world. We evaluate the following attack strengths: 1) Full Attack: deals 10 HP of damage, instantly killing any agent in the attack radius 2) Half Attack: deals 5 HP of damage 3) Quarter Attack: deals 2.5 HP of damage. The main **Ocean** experiments in Section 4.2 were run with the Full Attack setting.

We find that reducing the attack strength from Full Attack to Half Attack significantly reduces the effectiveness and frequency of attacks: attacks per agent and homicides per attack drop considerably (Figure 22). As a result, populations with lower attack strength grow larger, with agents moving and eating more frequently, resulting in greater biomass utilization. Further reducing strength to the Quarter Attack setting yields diminishing returns in terms of further reduction in violence and facilitation of additional population growth. Intuitively, when attacks do less damage, a homicide

requires more attacks per agent, reducing the lethality of each attack. However when the lethality is reduced, attacking behavior becomes less useful and therefore less likely to be selected for in the agent population. Agents with lower attack strengths therefore do not become as skillful at attacking and spend more time foraging for food.

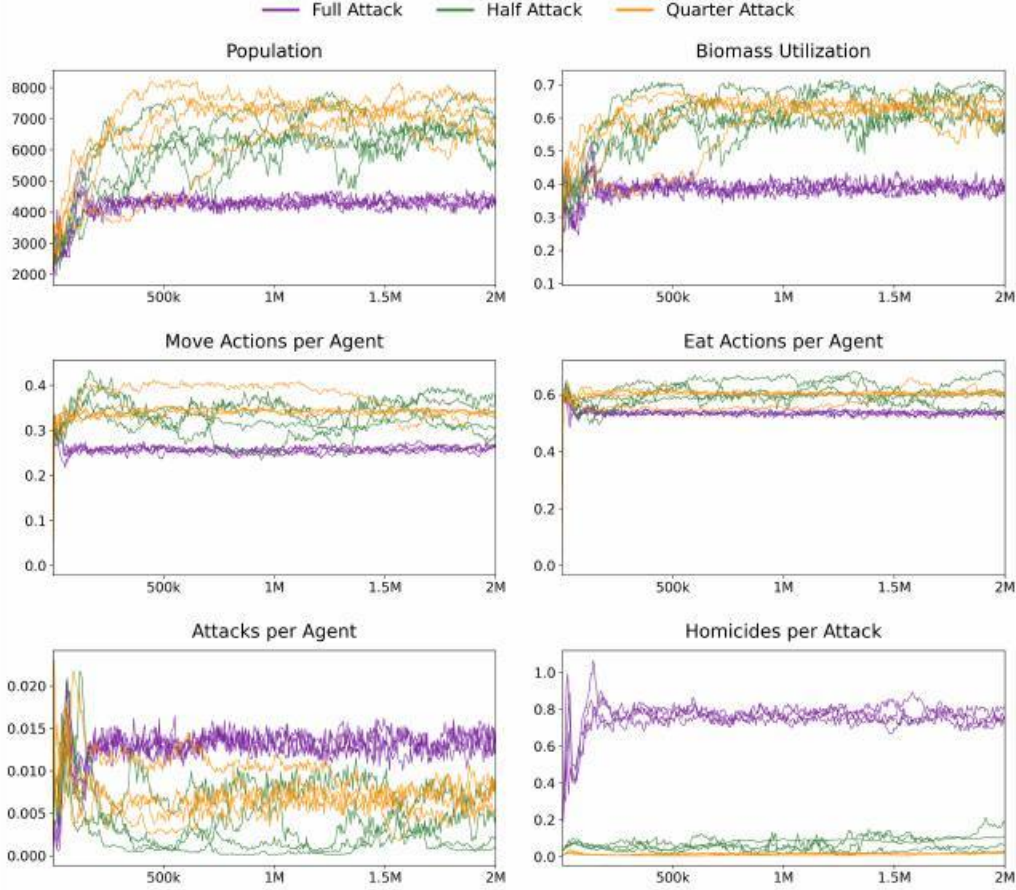


Figure 22: Population and action frequency data over 2M steps in a 256×256 **Ocean** world for (RCV+A) agents across Full Attack, Half Attack, and Quarter Attack strength settings. Four seeds are plotted per attack strength setting.

F.8 VARYING AGENT HP

We investigate the effect of reducing the starting HP of child agents and lowering the healing rate to apply pressure for agents to ‘grow’ over their lifetime in order to survive. The healing rate parameter determines how much HP an agent can regenerate per healing step, which converts available energy to HP. Agents can influence healing only by maintaining enough energy.

We study two settings: 1) Full HP: child HP starts at 5 and healing rate is 1 HP per 0.1 energy 2) Low HP: child HP starts at 1, with healing rate 0.5. Max agent HP is 10 in all settings. The main experiments (Section 4) were run with the Full HP setting.

Across the two settings, there is no significant difference in homicides per attack, eat action frequency, or biomass utilization (Figure 23). In the Full HP setting, populations grow slightly larger, while in the Low HP setting, agents attack less frequently and move slightly more. We hypothesize that agents born with lower HP must focus on foraging early on in their lifetime in order to grow to full size, which would explain this difference.

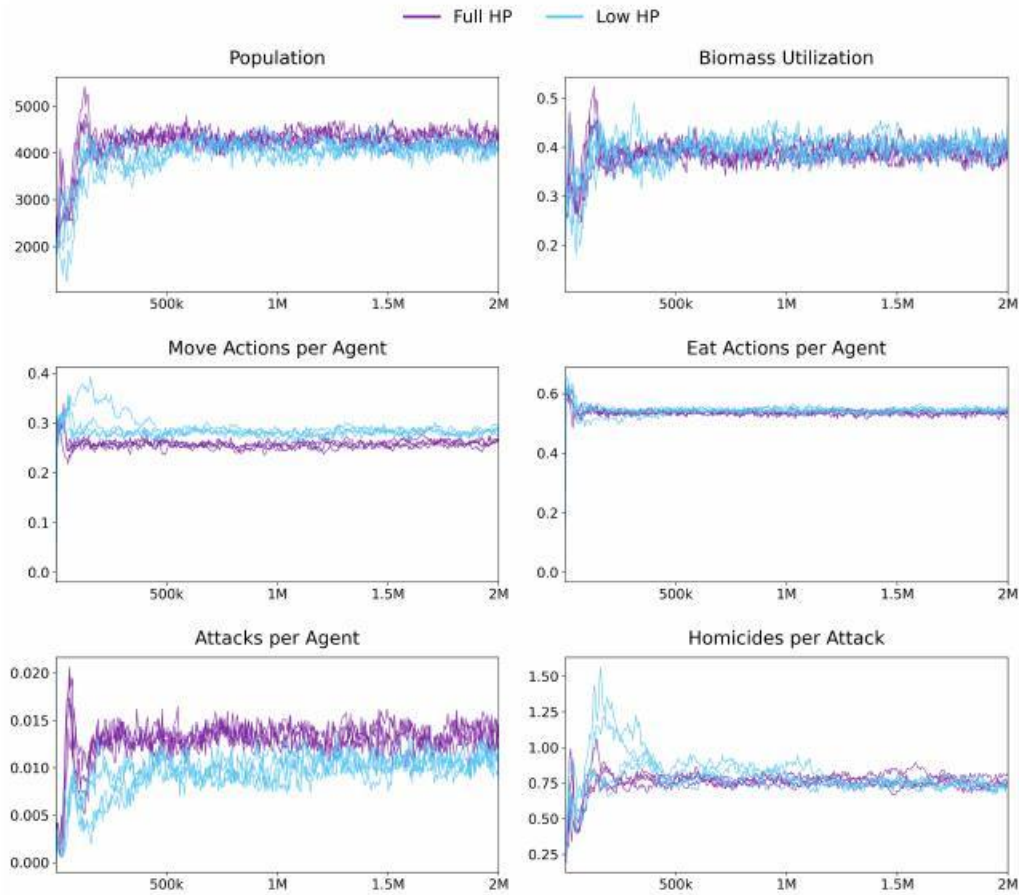


Figure 23: Population and action frequency data over 2M steps in the 256x256 Ocean world for (RCV+A) agents across Full HP and Low HP settings. Four seeds are plotted per HP setting.